Software Requirements

Handout # 4

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Recap of Last Three Lectures

- Kinds of requirements
  - Functional
  - Non-functional
  - Domain
  - Inverse
  - Design and implementation constraints

Topics Covered In This Lecture

- There also exists another view of requirements apart from different kinds of requirements we have studied so far.
  - Another view of requirements
- There are some problems which occur in requirements, that are necessary to be identified and properly attended.
  - Problems in requirements

Another View of Requirements

- In general requirements can be viewed as
  - User/customer requirements

    OR
  - System contract requirements

User/Customer Requirements

User/Customer Requirements - 1

- Functional and non-functional requirements should be stated in natural language with the help of forms or simple diagrams
- describing the expected services of a system by the User under certain constraints

User/Customer Requirements - 2

- These are understandable by users, who have no, or little, technical knowledge
- System design characteristics should be avoided as much as possible

User/Customer Requirements - 3

- It is a good practice to separate user requirements from more detailed system requirements in a requirements document

User/Customer Requirements - 4

- Including too much information in user requirements, constraints the system designers from coming up with creative solutions

User/Customer Requirements - 5

- The rationale associated with requirements is very important. It helps in managing changes to requirements
- Because Rational comes from the process it represents.

System Contract Requirements

System Contract Requirements - 1

- Sets out the system services and constraints in detail
- May serve as the basis of contract for implementation of the system
- Should be complete and consistent

System Contract Requirements - 2

- They are used by the designers and developers as the starting point for system design
- They should be understood by technical staff of the customer organization and the development team

System Contract Requirements - 3

- In principle, these requirements should also state ‘what’ the system does, rather than ‘how’ it is implemented
- However, with the level of details needed to specify the system completely, it is not possible to exclude all design information
  - E.g. Degree will be awarded once the CGPA is calculated from all the results of the passed courses in a Database

System Contract Requirements - 4

- An initial architecture of the system may be defined to help structure the requirements specification
- In most cases, systems interoperate with other systems
  - Use of specific design may be included as an external requirement

System Contract Requirements - 5

- Natural language is often used to describe system requirements
- Some specification languages may be used with natural language, which add structure to specifications and reduce ambiguity

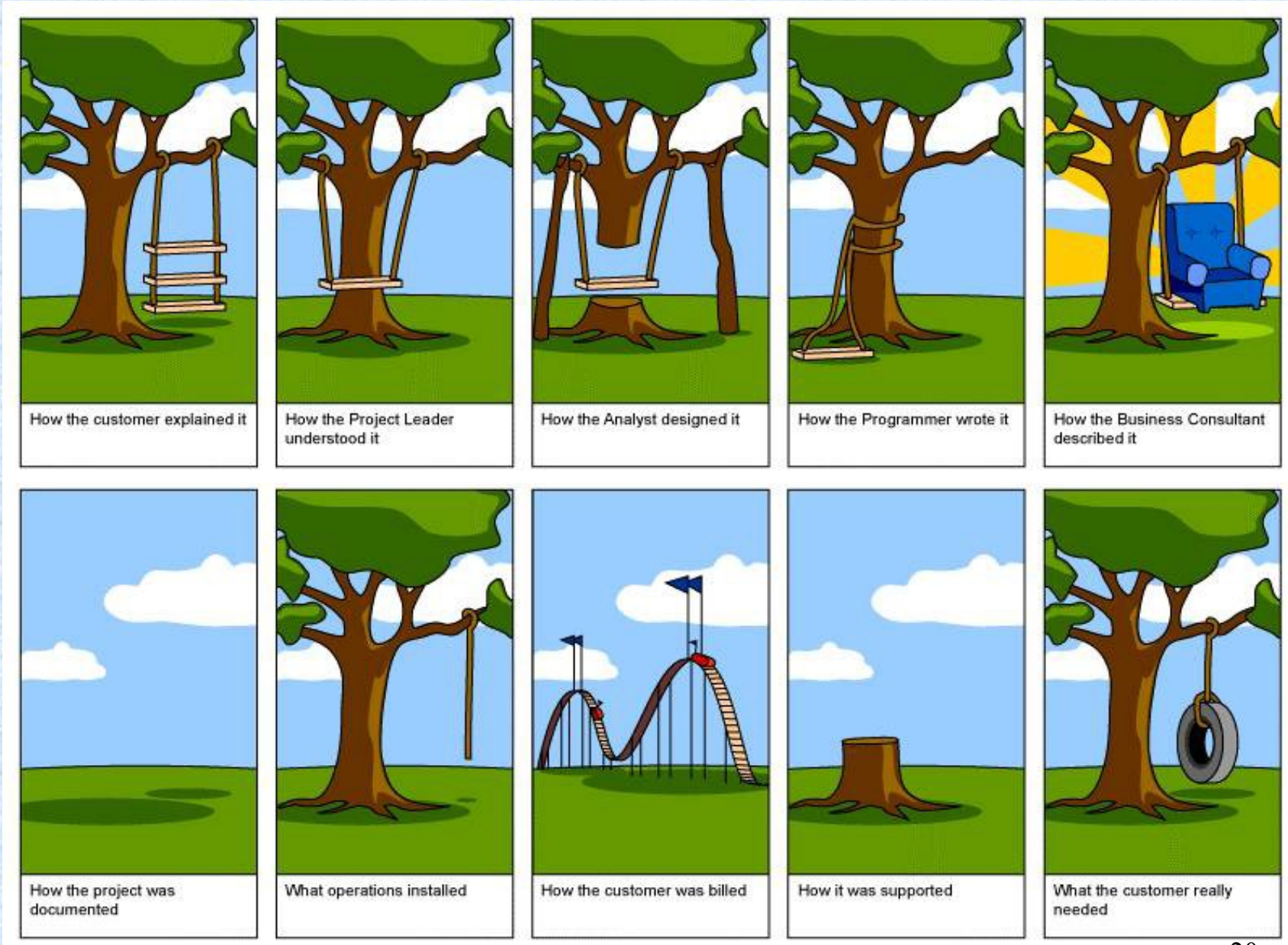
System Contract Requirements - 6

- Unified Modeling Language (UML) is a specification language, which has become the de-facto standard for modeling requirements

Requirements Problems

Requirements Problems - 1

- The requirements don't reflect the real needs of the customer for the system
- Requirements are inconsistent and/or incomplete
- It is expensive to make changes to requirements after they have been agreed upon (Who bears the costs of the change)


How the customer explained it
How the Project Leader understood it
How the Analyst designed it
How the Programmer wrote it
How the Business Consultant described it
How the project was documented
What operations installed
How the customer was billed
How it was supported
What the customer really needed

Requirements Problems - 2

- There are misunderstandings between customers, those developing the system requirements, and software engineers developing or maintaining the system
- The Four Worlds

Problems with Natural Languages - 1

Requirement specification in natural language pose some problems which include

- Lack of clarity
  - Jackson (Jackson, 1995) gives an excellent example of this when he discusses signs displayed by an escalator. These said 'Shoes must be worn' and 'Dogs must be carried'. *Work out the conflicting interpretations of these phrases.*
- Requirements confusion
  - For example: Full model of software quality consists of following **non-functional** qualities:
  - **Functional suitability**: Degree to which a product or system provides functions that meet stated and implied needs (**functional requirements**).
- Requirements amalgamation
  - Several different requirements may be expressed together. E.g. Time taken to download 2000 phone songs in Infotainment system after Bluetooth pairing and connection

Problems with Natural Languages - 2

- Natural language understanding relies on the specification readers and writers using the same words for same concept (E.g. Grades vs. Marks)
- A natural language requirements specification is over-flexible.

"You can say the same thing in completely different ways"

Problems with Natural Languages - 3

- It is not possible to modularize natural language requirements. It may be difficult to find all related requirements
  - To discover the impact of a change, every requirement have to be examined

    e.g. Admission is confirmed once the fee is paid

An Example

- Imagine the following example of an informal specification from a critical system [1] :
  - "The message must be triplicated. The three copies must be forwarded through three different physical channels. The receiver accepts the message on the basis of a two-out-of-three voting policy."
- Questions: Can you identify any ambiguities in this specification?

[1] - C. Ghezzi, M. Jazayeri, D. Mandrioli, "Fundamentals of Software Engineering", Prentice Hall, Second Edition, page 196 - 198

Answer:

- can a message be accepted as soon as we receive 2 out of 3 identical copies of message or do we need to wait for receipt of the 3rd?
- What if all three messages are different?

Impact of Wrong Requirements

- When requirements are wrong, systems are late, unreliable and don't meet customers needs
- This results in enormous loss of time, revenue, market share, and trust of customers

Summary

- Discussed requirements from the user/customer's perspective and also explored issues related to system contract requirements
- Discussed requirements problems

References

- 'Requirements Engineering: Processes and Techniques' by G. Kotonya and I. Sommerville, John Wiley & Sons, 1998
- Software Requirements: Objects, Functions, and States by A. Davis, PH, 1993
- Software Engineering 6th Edition, by I. Sommerville, 2000
- Software Engineering 5th Edition, by R. Pressman